MILITARY SPECIFICATION

RADAR SET AN/APS-82

This specification has been approved by the
Bureau of Aeronautics, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall be a 360° scan Airborne Early Warning and Control radar and shall provide means for searching for and detecting aircraft targets and in addition shall have means for determining height of said targets and directing an interceptor on a collision or off-set intercept course.

1.2 Classification - The Radar Set AN/APS-82 shall be of one type and shall consist of the following:

Units	Type Designation	Max. Allow Size (Inches) W	H	L	Max. Allow. Wt. Lbs.	Appl. Para.
Antenna Assembly	AS-(*)/APS-82	(17-1/2' x 4') reflector approximately			375	3.3.1
Waveguide Switch	SA-(*)/APS-82	8	13	11	15	3.3.2
Dummy Load	DA-(*)/APS-82	10	8	27	14	3.3.3
Bi-Directional Coupler	CU-(*)/APS-82	5	8	12	8	3.3.4
Duplexer Assembly	CU-(*)/APS-82	12	20	12	12	3.3.5
Modulator-Transmitter	T-(*)/APS-82	22	23	30	330	3.3.6
Receiver, Radar	R-(*)/APS-82	12	7	9	15	3.3.7
Uni-Directional Coupler	CU-(*)/APS-82	5	5	12	4	3.3.8
Mono-plexer Assembly	CU-(*)/APS-82	6	7	22	10	3.3.9
Indicators, Azimuth and Range (2)	IP-414/APA-125	19-1/2	12-3/16	24-1/2	212	3.3.10
Displays, Height Finder (2)	IP-(*)/APS-82	8	6	20	70	3.3.11
Controls, A.I.C. (2)	C-(*)/APS-82	19-1/2	3	7	20	3.3.12

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Units	Type Designation	Max. Allow Size (Inches) W	H	L	Allow. Wt. Lbs.	Appl. Para.
Computers, A.I.C. (2)	CP-(*)/APS-82	11	6	20	80	3.3.13
Amplifier, Electronic Control	AM-(*)/APS-82	8	9	14	25	3.3.14
Power Supply	PP-(*)/APS-82	13	11	19	55	3.3.15
Radar Set Control	C-(*)/APS-82	5-3/4	7-1/2	8	5	3.3.16
Maintenance Control	C-(*)/APS-82	12	8	7	15	3.3.17
Transformer Assy.	TF-(*)/APS-82	6	6	18	14	3.3.18
Air Compressor	HD-(*)/APS-82	10	5	10	12	3.3.19
Dehydrator	HD-(*)/APS-82	4	7	9	4	3.3.20
Synchronizer	SN-(*)/APS-82	12	8	19-3/4	30	3.3.21
Control Box, Relay Synchronizer	C-(*)/APS-82	5-3/4	1-7/8	6-1/4	1	3.3.22
Mountings						
Mounting Base (Power Supply	MT-(*)/APS-82					
Mounting Base (Electronic Control Amplifier)	MT-(*)/APS-82					
Mounting Base (Height Finder Display AIC Computer)	MT-(*)/APS-82					
Mounting Base (Radar Indicator)	MT-(*)/APS-82					
Mounting Base (Radar Receiver)	MT-(*)/APS-82					
Mounting Base (Maintenance Control)	MT-(*)/APS-82					
Mounting Base (Synchronizer, Radar Relay)	MT-(*)/APS-82					

(The weights of mountings are included in the equipment weight (see 3.2.4.4)

NOTE: The parenthesis (*), in the type designation throughout this specification, shall be replaced by either a number or letter furnished by the procuring agency upon application by the contractor in accordance with 3.2.15 after each contract has been awarded. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

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1.3 Accessories - The equipment shall operate with the following accessories, however, these accessories are not to be supplied as part of this equipment.

(1)	AN/APX-7	Radar Recognition Set
(2)	AN/ASA-13( )	Ground Position Indicator
(3)		Central Gyro Reference System
(4)		Power Dist. Panel
(5)	AN/ART-28( )	Radar Relay Link
(6)	AN/UPM-44B	Test Set, Radar
(7)		Off-Set Intercept Chart

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-E-1	Electron Tubes and Crystal Rectifiers
MIL-C-172	Cases: Bases, Mounting; and Mounts, Vibration (For Use with Electronic Equipment in Aircraft)
MIL-E-4682	Electron Tubes, Choice and Application of
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing; Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Limits, Tests, and Design Requirements, Aircraft Electrical and Electronic Equipment
MIL-C-6781	Control Panel; Aircraft Equipment Rack or Console-mounted
MIL-P-7788	Plate, Plastic, Cockpit and Interior Controls Lighting
MIL-E-7894	Electric Power, Aircraft Characteristics of
MIL-P-17555	Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts.
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment
MIL-T-18303	Test Procedures, Preproduction and Inspection, for Aircraft Electronic Equipment, Format for

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Military (Cont'd)

MIL-I-19165	Indicator Group AN/APA-125
MIL-F-25173	Fastener, Control Panel, Aircraft Equipment

Bureau of Aeronautics

EI-207	Control Panels; Electrical and Electronic Equipment, Rack or Console Mounted, Installation of
SAR-300	Engineering Change Proposals, Electronic Material Changes and Electronic Material Bulletins; Preparation of

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

Air Force - Navy Aeronautical

AND 10461	Connections - Input Power Airborne Electronic Equipment; 28-Volt DC, 115-Volt Single Phase AC, and/or 115/200 Volt 3-Phase AC

2.2. Availability of Documents:

(1) Refer to both the title and number, when requesting specifications, publications and drawings. All requests should be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable specifications, standards and publications may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, Philadelphia 11, Pennsylvania. Attention: Code ODPT.

(3) Copies of applicable drawings may be obtained upon application to the Bureau of Aeronautics, Navy Department, Washington 25, D.C., Attention: Avionics Division

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2.3 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring agency.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing, the following shall govern:

(1) Parts and Materials as approved by Specification MIL-E-5400 shall be given first consideration.

(2) When the contractors can demonstrate that the use of standard MIL parts or materials will not fulfill the design objectives because of size, weight, performance or other reasons, materials and parts shall be used which most nearly meet or exceed the requirements of the respective standard specifications.

(3) When a non-standard part is used, the meeting of general material requirements such as non-inflammability, fungus resistance, non-toxicity, etc., and the meeting of environmental conditions as required by Specification MIL-E-5400 or MIL-T-5422 shall be of prime consideration.

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(4) In the selection of parts, the above considerations shall apply if the assembly is to be built for detail part maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed and the quality of the individual parts shall be such that the unit as a whole will pass material, electrical and environmental tests. Any assembly built to be expended shall be approved first by the procuring agency.

3.1.1 Non-Standard Part Approval and Material Approval - Approval for the use of non-standard, parts and materials shall be obtained as outlined in Specification MIL-E-5400 except as follows:

(1) Mechanical parts are expected to meet the material requirements of MIL-E-5400, and shall be submitted as non-standard parts only when the material requirements are not met.

(2) Non-standard parts, as defined in specification MIL-E-5400 of an electrical nature must be approved before the preproduction model of the equipment is delivered to the Bureau of Aeronautics. The request for approval must be forwarded by the contractor before the part is purchased or fabricated. In addition to, or in lieu of the NON-Standard Part Data Sheet the contractor is requested to submit itemized lists of the parts, their sources, and the contractor's drawing or purchase specification covering the parts.

3.1.2 Choice of Tube Types - Electron tubes used in the equipment shall be chosen in accordance with the following requirements and in the sequence noted:

(1) MIL-E-4682

(2) The use of any tube not meeting the requirements of (1) above must be approved by the Bureau of Aeronautics, however, before consideration for approval, the contractor must submit the following information:

(a) Quantitative data on the electrical advantages and mechanical adequacy of the proposed tube over the most nearly equivalent type on the acceptable listings.

(b) Test data and performance limits of the tube which must be maintained to insure satisfactory and reliable performance in the equipment.

(c) Information as to whether the tube manufacturer has received Military or Navy qualification approval for the particular tube.

3.1.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives to be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the Bureau of Aeronautics for consideration. Each request should be accompanied by complete supporting information.

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3.2 DESIGN, CONSTRUCTION AND PERFORMANCE:

3.2.1 General - This equipment shall fulfill all applicable design and performance requirements of Specification MIL-E-5400 for Class 1 equipment, unless otherwise specified herein. Unit mountings shall be in accordance with MIL-C-172.

3.2.2 Standard Conditions - The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests on the equipment.

Temperature	Room ambient 25°C (+15°C or -5°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input power voltage	Three phase 4 wire system 115 volts ±1.0 VAC line to neutral and 27.5 volts ±0.5 VDC.

3.2.3 Service Conditions - The equipment shall operate satisfactorily under any of the service conditions or reasonable combinations of service conditions as specified in MIL-E-5400, unless otherwise specified herein.

3.2.3.1 Sand and Dust - The equipment shall be designed to be as dustproof as practicable consistent with cooling requirements. Testing of the equipment in a turbulent atmosphere filled with sand and dust concentration will not be required of the contractor.

3.2.3.2 Vibration - The vibration requirement shall apply except the double amplitude value of 0.06 inch shall be changed to 0.04 inch.

3.2.3.3 Altitude - Continuous operation at barometric pressures ranging from 30 in. Hg down to 8.8 in. Hg (approximately 30,000 ft. altitude). The pressure remaining constant for long periods of time or varying at a rate as high as 0.5 in. Hg per second.

3.2.4 Total Weight - The total weight of the equipment excluding cables, shall be a minimum consistent with good design and shall not exceed 1,350 pounds.

3.2.5 Operational Stability - The equipment shall operate with optimum performance for 24 hours, continuously or intermittently, without necessity for readjustment of any controls which are non-accessible during flight.

3.2.6 Operating Life

3.2.6.1 Reliable Operating Life - The equipment shall have a reliable operating life of at least 500 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer.

3.2.6.2 Total Operating Life - The equipment shall have a minimum total operating life of 2000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

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3.2.7 Interconnection Cabling - This equipment shall be capable of satisfactory operating using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded except that a minimum number of the individual wires may be shielded, when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished.

3.2.8 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with the requirements of Specification MIL-E-5400. However, cables and that portion of the connectors which is to be attached to the cables shall not be supplied as part of the equipment.

3.2.9 Console Controls - The Radar Set Control shall be satisfactory for mounting in a console arrangement when used in aircraft. This unit shall meet the requirements of specification MIL-C-6781, shall have a plastic lighting plate meeting the requirements of specification MIL-P-7788 and shall be equipped with fasteners meeting the requirements of specification MIL-F-25173 for mounting in accordance with requirements of specification EI-207. All remote control units shall meet the requirements of MIL-C-6781 and MIL-P-7788. The lamps shall be connected in parallel with one side grounded to the chassis of the unit and the other side being connected to a separate contact, in the electrical connector receptacle of the unit, for connection to an external 28V input lead. The brilliance of the lamps will be controlled externally and the brilliance control shall not be part of the control unit, except the IP-414/APA-125 shall be as described in MIL-I-19165. The arrangement of controls and all lettering and markings proposed for use on the front panel of control units shall be approved by the Bureau of Aeronautics prior to fabrication of any complete units, and control knobs shall be shape and color coded.

3.2.10 Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-E-7894 and shall give specified performance from the following power sources with characteristics as defined in MIL-E-7894 having limits as specified herein. The power required shall not exceed the specified amounts.

3.2.10.1 AC Power - The AC primary input power to this equipment shall be of the 115/200 volt constant frequency type, shall not exceed 8000 VA and shall have the following characteristics:

(1) Operating frequency may vary between 380 and 420 cycles per second.

(2) A 3-phase, 4-wire grounded neutral system shall be used.

(3) Whenever 3-phase input power is averaged, as in 3-phase rectification circuits, the bus voltage variation shall be as if all three-phase to neutral voltages are within the limits of 113 to 117 volts. Bus voltage variation for single-phase loads shall be within the limits of 112 to 118 volts. The equipment shall operate at reduced performance between the extreme limits of 108 to 112 volts and 118 to 121 volts.

(4) Line voltage drops as specified in MIL-W-5088 may be expected. AN/APS-82 input power transformers may be tapped if necessary to permit the installing activity to make compensation adjustments for the line voltage drops between the point of primary power regulation and the equipment input.

(5) Transients associated with the AC primary input power may vary in amplitude from 60 to 180 volts but shall never exceed 0.2 seconds.

3.2.10.2 DC Power 28-Volt System -

Operating Voltage Limits:	25 to 29 volts
Current:	20 Amps.

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3.2.11 Input Power Connection - Input power connections shall be in accordance with drawing AND-10461, where applicable.

3.2.12 Overload Protection - In general primary power overload protection shall be provided external to the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall have suitable protective devices, either within or outside the equipment, or shall be proportioned to withstand such overload without permanent damage to the equipment. The use of protective devices within the equipment shall be held to a minimum. There shall be no fuses in the equipment.

3.2.12.1 Under Voltage Protection - The equipment shall not be damaged by voltages below the minimum specified and shall automatically resume normal operation when the voltage returns within limits.

3.2.13 Standby Provisions - The equipment shall provide for a standby position wherein rated voltage shall be applied to all circuits except that high power shall not be applied to the modulator and transmitter.

3.2.14 Moisture and Fungus Resistant Treatment - Equipment shall be fungus proofed by selection of parts and materials as non-nutrient for fungus or the parts and materials shall be treated prior to their use in the equipment so that overall spraying of the equipment is not necessary.

3.2.15 Nomenclature and Nameplates - Nomenclature assignment and Nameplate approval for equipment identification shall be in accordance with the requirements of Specification MIL-N-18307.

3.2.16 Interchangeability - The interchangeability requirements between like assemblies, subassemblies and replaceable parts as specified in MIL-E-5400 shall be extended to require interchangeability of units of this equipment with the latest model of any previously produced equipment. Such interchangeability shall be measured against a model, manufacturing drawings, or other technical information as provided by the Bureau of Aeronautics for this purpose. In the event that requirements of MIL-E-5400 conflict with the requirements for interchangeability, the interchangeability requirements shall govern.

3.2.17 Provisions for Maintenance and Field Testing - The equipment shall provide for ease of maintenance to the greatest extent practicable. Provisions for field testing shall be as specified in MIL-E-5400. The following features shall be design goals for the equipment:

(1) A minimum number of servicing and maintenance adjustments.

(2) All maintenance to be available from the outside of the units without removal of dust covers.

(3) Interchangeability of major system units without requiring readjusting maintenance controls.

3.2.18 Radio Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.2.19 Performance - The equipment shall be designed as part of the Airborne Early Warning and Control System and shall be capable of:

(1) Transmitting and receiving radar signals and of determining target height.

(2) Furnishing north-stabilized azimuth-synchro, trigger, radar and IFF video information to the radar relay synchronizer, SN-(*)/APS-82.

(3) Controlling air to air interceptions.

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3.2.19.1 Warm-Up Time - The equipment shall be stabilized and capable of the specified performance after 15 minutes of warm-up time.

3.2.19.2 Frequency Characteristics - The radar equipment shall be capable of operating at a fixed and/or preset frequency within the 2850 to 2910 mcs range. The IFF Parts of the antenna and rotary joint shall operate within the frequency range of 1010 to 1110 mc/s.

3.2.19.3 Modulation Characteristics - The transmitter shall be pulse modulated at a pulse repetition rate of 300 ±30 cps by a pulse width of 2 ∠ 0.1 microseconds.

3.2.19.4 Transmitter Power Output - The equipment shall be capable of delivering 1 megawatt nominal but not less than 0.65 megawatt peak measured at the directional coupler.

3.2.19.5 System Noise Figure and Minimum Perceptible Signals - The overall system noise figure shall be less than 8 db measured at the sum and difference channels near their respective directional coupler guides. This measurement shall take precedence over sensitivity when determining performance of system. The radar minimum perceptible signal shall be greater than 136 db below one watt in the sum channel.

3.2.19.6 Anti-Jam Provisions - The following anti-jam provisions shall be incorporated in the receiving system:

(a) Sensitivity Time Control

(b) IAGC controlled IF strip

3.2.19.7 Indication -

The indicating system consists of two general purpose 10 inch Range and Azimuth Indicators and two Height Indicators with type "A" presentation. Search Radar data is presented on the Range and Azimuth Indicators and in addition Time Shared Cursors and Strobes are presented to aid in semi-automatic computation of collision or off-set intercepts. The "sum" and "monopulse" videos are presented on the Height Finder Indicators to permit accurate aiming of the radar beam at the target.

3.2.19.7.1 Range and Azimuth Indicator - The range and azimuth indicator shall be an IP-414/APA-125.

3.2.19.7.2 Height Indicator - The height indicator shall display the monopulse video signal received by the radar. Semi-automatic computation of the height shall be incorporated in this unit.

3.2.19.8 IFF Provisions -

(1) Display - The display of IFF Video on that indicator as well as mode of challenge shall be selected by the operation of one of the three IFF selector switches on the panel of the IP-414/APA-125 Indicator.

(2) Synchronization - Synchronization of the accessory IFF equipment shall be accomplished by the SN-(*)/APS-82 or an auxiliary unit which is used only if the SN-(*)/APS-82 is removed.

(3) RF Provisions - The AN/APS-82 radar set and the accessory AN/APX-7 IFF equipment shall share a common rotary joint and antenna unit.

3.2.19.9 VSWR - The VSWR between 2880 ±30 mc/s measured at the input to the Duplexer with the antenna looking into free space shall be no greater than 1.35 to 1.

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3.2.19.10 Height Finder Accuracy - The height finding system shall be capable of measuring the target elevation to an accuracy of ±1.25 degrees.

3.2.19.11 System Pressure - All units in the system requiring pressurization shall be capable of holding pressure of 5 pounds per square inch above the maximum operating pressure.

3.2.19.12 Workmanship - Workmanship, fabrication and manufacturing process shall conform to the requirements of MIL-E-5400.

3.3 Detail Requirements

3.3.1 Antenna Assembly Unit - AS-(*)/APS-82

3.3.1.1 Function - The antenna assembly shall provide for radiating and receiving radar and IFF signals. It shall provide means for varying and reporting the azimuthal and vertical position of the radar beam to the radar. Tilt stabilization shall be accomplished by automatic positioning of the reflector with respect to the feed. Provision shall be made for manual tilt control and antenna speed control.

3.3.1.2 Form Factor and Mounting - The dimensions of the antenna assembly shall be such as to permit mounting and operation within the radome mounted on top of the WF-2 aircraft. (See 1.2)

3.3.1.3 Stabilized Operation - Under stabilized operation the tilt angle (elevation angle of the null of the difference pattern) shall be stabilized with respect to the gyro indicated horizontal plane with an accuracy of ± 0.75 degree at zero degrees stabilized tilt with a linear change to ±1 degree at ±14 degrees stabilized tilt. These accuracies shall be maintained for pitch and roll and manual displacements resulting in tilt angles of ±14 degrees, tilt rates of 30 degrees/sec or less or tilt accelerations of 20 degrees/sec/sec or less. These accuracies shall be maintained in the absence of the radome.

3.3.1.4 Tilt Control - Throughout 360° of rotation it shall be possible to manually tilt the axis of the radar beam through an angle of ±14 degrees, with respect to the mounting plane.

3.3.1.4.1 Contents of Unit - The antenna unit shall provide the following:

(1) Tilt Actuator Mechanism

(2) Reflector

(3) IFF Dipole array

(4) Split feed horn

(5) Rotary Joint

(6) Azimuth drive and gear box assembly

(7) Resolver isolation amplifiers

(8) Heading information components

3.3.1.5 Tilt Limit - When combined tilt angles exceed ±14°, the equipment shall not be damaged and shall return to satisfactory operation when the aircraft returns within the stabilization limits. Electrical limit controls shall be provided to restrict the tilt operation when beyond ±14 degrees. Mechanical limit controls shall also be provided to prevent damage to the equipment and/or radome in the event of failure of the electrical limit controls.

3.3.1.5.1 Tilt Dial - A calibrated tilt dial shall be provided which indicates the elevation of the radar beam with respect to the mounting plane with an accuracy of ±0.5 degree.

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3.3.1.6 Tilt Resolver - A synchro resolver shall be attached to the antenna whose rotation shall correspond to the tilt angle of the beam.

3.3.1.7 Isolation Amplifier - A two channel isolation amplifier shall be provided to drive the tilt resolver.

3.3.1.8 Tilt Motor - A low inertia servo motor shall be provided to tilt the reflector. The motor shall receive control voltages from the Electronic Control Amplifier AM-(*)/APS-82. (See 3.3.14).

3.3.1.9 Frequency Characteristics - The IFF Antenna section shall perform as specified herein without adjustment throughout the frequency range of 1010 to 1110 Mc. The radar antenna section shall perform as specified herein without adjustment throughout the frequency range from 2850 to 2910 Mc.

3.3.1.10 Antenna Gain - The radar antenna shall have a gain relative to an isotropic radiator of at least 35 db measured at zero tilt in the sum-channel at the stationary end of the rotary joint. A minimum gain of 32.5 db shall occur at ±14 degrees beam tilt. The IFF antenna gain at zero tilt shall be at least 25 db relative to an isotropic radiator and shall be at least 23 db at ±14° beam tilt.

3.3.1.11 Radar Sum Radiation Patterns (One-Way)

3.3.1.11.1 Polarization - The radiated field shall be horizontally polarized.

3.3.1.11.2 Horizontal Pattern - The radiation pattern in the horizontal plane at zero tilt shall not be greater than 1.55 degrees between the half power points. At ±14 degrees beam tilt, the beam width shall not exceed 2.1 degrees between the half-power points.

3.3.1.11.3 Vertical Pattern - The radiation pattern in the vertical plane shall not be greater than 6.2 degrees between the half-power points.

3.3.1.11.4 Side Lobes - The side lobes of both vertical and horizontal patterns at zero tilt shall be at least 20 db below the main lobe. At ±14 degree beam tilt, the side lobes of both vertical and horizontal patterns shall be at least 18 db below the main lobe.

3.3.1.12 Radar Difference Radiation Pattern - The radar difference pattern shall consist of two lobes of opposite polarity one above the other, with a null between them. The null shall be located at the same elevation angle as the peak of the sum-channel main lobe within ±1.0 degrees.

3.3.1.13 IFF Radiation Pattern (One-Way) -

3.3.1.13.1 Polarization - The radiated field shall be vertically polarized.

3.3.1.13.2 Horizontal Pattern - The radiation pattern in the horizontal plane shall be not more than 6 degrees between one-half power points.

3.3.1.13.3 Vertical Pattern - The radiation pattern in the vertical plane shall be not more than 20 degrees at one-half power points.

3.3.1.13.4 IFF and Radar Beam Alignment - In the horizontal plane, the maximum deviation between the axis of the radar sum main lobe and the IFF main lobe shall not exceed 0.5 degrees. In the vertical plane, the main lobe of the IFF radiation pattern shall enclose the main lobe of the radar sum radiation pattern.

3.3.1.13.5 Side Lobes - The side lobes in the vertical plane shall be at least 15 db below the main lobe. The side lobes in the horizontal plane shall be at least 18 db below the main lobe at zero tilt. For all other beam tilts the azimuth side lobes shall be at least 17 db below the main lobe.

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3.3.1.14 Interference between Radar and IFF Equipment - There shall be no coupling between the IFF system at the IFF frequency and the radar system which will produce a detectable signal in the Receiver R-(*) /APS-82. For signals in frequency range 2850 to 2910 megacycles, there shall be a minimum of 40 db rejection between the IFF connector at the bottom of the rotary joint and any channel at the top of the rotary joint (IFF connector, sum-waveguide, difference-waveguide). The feedhorn must be removed to carry out this test.

3.3.1.15 RF Power - The entire assembly shall be capable of handling one megawatt of r.f. power in the frequency range 2880 ±30 Mc applied to the sum-channel at the lower end of the rotary joint. The entire assembly shall be capable of handling 2 kilowatts of r.f. power in the frequency ranges (1010-1110 Mc) applied at the IFF cable connector at the bottom of the rotary joint. Both of the above requirements shall be met with a maximum of 1 atmosphere of dry air in the waveguide system and with an external pressure corresponding to 30,000 foot altitude.

3.3.1.16 Rotating Joint - This unit shall contain the parts necessary for the transfer of IFF and radar energy and control signal between the rotating antenna assembly and the stationary transmission lines running to the radar and IFF equipments.

3.3.1.16.1 Attenuation - The attenuation (insertion loss between matched terminations) of the radar sum-channel shall not exceed 0.3 db. The attenuation of the IFF channel shall not exceed 0.5 db. These requirements apply to the rotary joint alone. The attenuation of the difference channel shall not exceed 1.0 db.

3.3.1.16.2 Electrical Connections - There shall be separate connections for both channels of the radar and the IFF equipment.

3.3.1.17 Voltage Standing Wave Ratio - With the antenna looking into free space the voltage standing wave ratios measured at the bottom (stationary) end of the rotary joint shall not exceed the following limits:

3.3.1.17.1 Radar Sum Channel - 1.3:1 in the frequency range 2850 to 2910 megacycles.

3.3.1.17.2 Radar Difference Channel - 2.0:1 in the frequency range 2850 to 2910 megacycles.

3.3.1.17.3 IFF Channel - 2.0:1 in the frequency range 1010 to 1110 megacycles.

3.3.1.18 Azimuth Drive Assembly -

3.3.1.18.1 Speed - During normal search, the scanning drive mechanism shall be capable of continuously rotating the beam through 360 degrees at a rate of 6 rpm ±20%.

3.3.1.18.2 Height Finding Speed - During height finding the scanning drive mechanism shall be capable of rotating the antenna past the desired target at 1 rpm ±20%.

3.3.1.18.3 Acceleration - The scanning drive mechanism shall be capable of accelerating the antenna from 1 rpm to 6 rpm in less than two seconds.

3.3.1.18.4 Deceleration - The scanning drive mechanism shall meet the same deceleration requirements as the acceleration requirements listed above.

3.3.1.18.5 Magnetic Heading Indicator - A "20 power" differential generator shall be provided which will be driven by the antenna assembly at 1:1 speed ratio to provide azimuth magnetic heading information to the radar and Relay Synchronizer Control Box C-(*)/APS-82.

3.3.1.18.6 Tilt Information - A synchro resolver driven at 1:1 speed ratio by the antenna assembly shall be provided to furnish azimuth information for computing the tilt angle.

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3.3.1.18.7 Resolver Isolation Amplifier - A two channel isolation amplifier shall be provided to drive the azimuth resolver.

3.3.1.18.8 Heading Marker Indicator - A switch and cam assembly shall be incorporated to provide a trigger for the heading marker generator when the antenna is trained at 0 degree ±0.5 degree relative bearing. Zero degree relative bearing shall be clearly marked on the pedestal.

3.3.1.19 Ease of Maintenance - Maintenance adjustments and the requirements for special tools or jigs to align the antenna pedestal shall be held to a minimum.

3.3.2 Waveguide Switch SA-(*)/APS-82

3.3.2.1 Function - The waveguide switch shall connect the radar sum-channel to the dummy load (3.3.3) when the actuator of the switch is de-energized. When the actuator is energized the switch shall connect the radar sum-channel to the antenna.

3.3.2.2 Electrical Connections - The following electrical connections shall be provided:

(1) Waveguide to Bi-Directional Coupler

(2) Waveguide to Antenna

(3) Waveguide to Dummy Load

(4) Control Wire - +28 volts

(5) Control Wire - Antenna Return

(6) Control Wire - Dummy Load Return

3.3.2.3 VSWR - The VSWR shall not exceed 1.12 over the frequency range of 2850 to 2910 mc/s.

3.3.2.4 Isolation - With the radar sum-channel switched to the dummy load, the isolation between the sum-channel input and the antenna connection shall be at least 50 db.

3.3.3 Dummy Load, DA-(*)/APS-82 -

3.3.3.1 Function - The dummy load shall be capable of providing a termination for the transmitter when the sum-channel is switched in the dummy load position.

3.3.3.2 Power - The Dummy Load shall be capable of absorbing 1.2 megawatt peak power and 700 watts average power.

3.3.3.3 VSWR - The VSWR shall not exceed 1.12 over the frequency range of 2850 to 2910 mc/s.

3.3.3.4. Electrical Connections - The dummy load shall be connected to the transmitter side of the sum-channel by means of the dummy load termination of the waveguide switch.

3.3.4 Bi-Directional Coupler, CU-(*)/APS-82 -

3.3.4.1 Function - The Bi-Directional Coupler shall provide for coupling a fixed percentage of the transmitter power into an auxiliary line. Also, it shall provide for coupling a fixed percentage of the auxiliary line power into the sum-channel (directed toward the antenna).

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3.3.4.2 Electrical Connections - The following electrical connections shall be provided:

(1) Waveguide to Duplexer

(2) Waveguide to Antenna

(3) Two Test Jacks

3.3.4.3 VSWR - The VSWR of the Bi-Directional Coupler shall not exceed 1.12.

3.3.4.4 Coupling - The coupling in the forward direction and the reverse direction shall be down 40 ±2 db, and each coupling output jack shall be marked with the attenuation to the nearest 0.1 db. The coupling shall be flat within ±0.25 db from 2850 to 2910 mc/s.

3.3.4.5 Directivity - The directivity shall be 40 db minimum.

3.3.5 Duplexer Assembly, CU-(*)/APS-82

3.3.5.1 Function - The duplexer assembly shall interconnect the transmitter, antenna, sum-channel and sum-channel receiver.

3.3.5.2 Contents of Unit - The duplexer assembly shall consist of the following parts:

3.3.5.2.1 Main Waveguide - The main waveguide shall contain mounting provisions for TR tube, ATR tube(s), and, AFC mixer. The main waveguide shall contain suitable flanges for connection between the flexible guide from the transmitter and the rigid guide to the antenna.

3.3.5.2.2 TR Tube - The TR tube shall be a broadband waveguide type such as the 1B58.

3.3.5.2.3 ATR Tube(s) - The ATR tube(s) shall be of the broadband type such as the 1B56.

3.3.5.2.4 Image Rejection Filter - The input of the image rejection filter (if used) shall be coupled to the output of the T.R. tube. The image rejection filter shall consist of a sharply-tuned wave trap. The resonant frequency of the trap shall be adjustable by means of a calibrated control. It shall be possible to set the trap frequency to any frequency in the range 2910 megacycles to 2970 megacycles. The insertion loss between matched waveguide terminations at the resonant frequency shall be greater than 17 db over a 2 megacycle bandwidth. The trap shall introduce a standing wave ratio of less than 1 db at a frequency 60 megacycles below the trap resonant frequency. Under all anticipated operating conditions the drift in tuning shall not exceed ±1 megacycle.

3.3.5.2.5 Balanced Mixer - The balanced mixer shall be capable of being coupled to the output of the TR tube or to the output of the image rejection filter. The local oscillator input to the mixer shall be a BNC receptacle. At rated crystal current the mixer and the associated BNC plug shall introduce a standing wave ratio of less than 6 db on the local oscillator cable.

3.3.5.2.6 I.F. Preamplifier - The IF preamplifier shall be mounted on the balanced mixer and shall be connected electrically to the output of the balanced mixer. The center frequency of the preamplifier shall be 30 ±0.25 megacycles and the bandwidth between half-power points shall be 2.4 ±0.2 megacycles. The noise figure of the preamplifier shall not be greater than 1.3 db. The preamplifier shall be capable of receiving STC and/or gated AGC voltages from the receiver. The preamplifier shall have outputs for monitoring both crystal currents individually.

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3.3.5.2.7 AFC Mixer - The AFC mixer shall mount on the main waveguide and shall couple a sample of the transmitter pulse to the AFC mixer crystal. The attenuation between the waveguide and the AFC crystal shall be 77 ± 3 db. The AFC mixer shall contain a crystal-current-adjusting means. Coaxial receptacles shall be provided for local oscillator input and IF output.

3.3.5.3 Controls - The following controls shall be incorporated in the duplexer assembly as described in 3.3.5.2.

(1) IMAGE REJECTION TUNE (if used)

(2) AFC CRYSTAL CURRENT (may be one or two controls)

3.3.5.4 Transmit Loss - The duplexer transmit loss shall not exceed 1.0 db.

3.3.5.5 Transmit VSWR - The duplexer transmit VSWR shall not exceed 1.25 over the band of 2880 ±30 mc/s.

3.3.5.6 Receive Loss - The duplexer receive loss exclusive of mixer, shall not exceed 1.5 db.

3.3.5.7 Receive VSWR - The duplexer receive VSWR shall not exceed 2:1.

3.3.5.8 Electrical Connections - The following electrical connections shall be provided as described in 3.3.5.2:

(1) WAVEGUIDE TO TRANSMITTER

(2) WAVEGUIDE TO BI-DIRECTIONAL COUPLER

(3) KEEP-ALIVE

(4) I.F. OUTPUT

(5) CRYSTAL #1 CURRENT

(6) CRYSTAL #2 CURRENT

(7) GROUND

(8) HEATER

(9) B+

(10) STC

(11) AGC

3.3.6 Modulator Transmitter, T-(*)/APS-82 -

3.3.6.1 Function - The Modulator Transmitter shall generate the high-level R.F. pulse and contain the high-voltage power supply, the pulse modulator, the magnetron, and the trigger generator.

3.3.6.2 High Voltage Power Supply - The high voltage power supply of the radar transmitter shall receive three-phase variable autotransformer voltages from the Transformer Assembly, transform, rectify and filter these voltages to provide dc voltage to the modulator section. The characteristics of the output shall be:

Nominal rating:	8 KV at 200 ma
Peak to Peak ripple:	1% maximum

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3.3.6.3 Modulator - The modulator shall generate the modulating pulse for the magnetron. It shall be triggered by a positive pulse of minimum amplitude 50 volts derived from the modulator trigger subchassis or external source. The pulse amplitude, rise and fall time of the output pulse shall be consistent with the requirements of the transmitter output tube.

3.3.6.4 Magnetron - The magnetron output shall have a current pulse width of 1.9 to 2.1 microseconds measured at 50% amplitude.

3.3.6.5 Power Output - The peak power of the transmitted pulse shall be determined by measuring the average power absorption of a matched water load and dividing by the repetition rate times the measured current pulse width at the 50% point. Minimum acceptable peak power output shall be 0.8 megawatts, measured directly at the output coupling of the transmitter.

3.3.6.6 Pulse Stability - Delay between the main trigger and the transmitter pulse shall not vary by more than .02 microsecond from pulse to pulse.

3.3.6.7 Transmittor Test Pulse - An auxiliary winding shall furnish a positive test trigger coincident with the transmitted pulse. The nominal amplitude of this trigger at the output of the transmitter shall be 40 volts.

3.3.6.8 Frequency Spectrum - The frequency spectrum of the transmitted energy shall be in accordance with Specification MIL-E-1 for the magnetron used (see 3.3.6.4). The bandwidth of the main lobe shall not exceed 1.5 mc/s at the 6 db points and all side lobes shall be at least 8 db below the main lobe.

3.3.6.9 Pressurization - The unit shall be capable of operating when pressurized at a minimum of 16 psi absolute.

3.3.6.10 Protection - The unit shall be provided with a safety interlock that will remove all high voltages when the "hat" is removed. Automatic protective circuits against high voltage power supply overcurrent and excess reverse diode current shall be incorporated.

3.3.6.11 Trigger Generator Output - A system output trigger coincident with the modulator trigger shall be provided and be of positive polarity and have a minimum peak amplitude of 35 volts across 1600 μμf. Its repetition rate shall be fixed within the range 300 ±30 cps. It shall be used to trigger the radar relay synchronizer, the indicators, the radar receiver, the height finder displays, the AIC computer and shall be furnished to the maintenance unit for test purposes.

3.3.6.12 Load Isolator - The load isolator shall reduce the power reflections to the magnetron by at least 10 db, and its insertion loss shall not exceed 0.5 db.

3.3.7 Radar Receiver, R-(*) /APS-82

3.3.7.1 Function - The receiver shall amplify the sum-channel and difference-channel signals from the IF pre-amplifiers and shall provide local oscillator and T-R voltages.

3.3.7.2 I.F. Subassembly - One each IF subassembly shall be supplied for the sum-channel and the difference-channel. Each I.F. subassembly shall consist of an IF amplifier and shall be controllable by IAGC. (See 3.3.7.2.3)

3.3.7.2.1 I.F. Amplifier Frequency - The IF amplifier shall be tuned to 30 ±0.25 mcs.

3.3.7.2.2 I.F. Bandwidth - The bandwidth of the IF subassembly shall be between 1 and 1.4 mcs measured between half-power points.

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3.3.7.2.3 IAGC Characteristics - IAGC shall be provided such that it shall be possible to compress a 10 microsecond signal 80 db above the noise to 1/2 of the amplitude of the initial spike. At this IAGC adjustment, it shall be possible to set the noise amplitude to a minimum of 1/2 of the amplitude of the spike.

3.3.7.2.4 Stability - There shall be no evidence of instability (regeneration) in the band-pass curve.

3.3.7.2.5 Output - Two low impedance outputs of the amplified IF signal shall be provided on each IF subassembly.

3.3.7.3 Detector-Video Amplifier Subassembly - The detector-video amplifier subassembly shall be capable of receiving the IF signal from the sum-channel and difference-channel IF amplifiers and supply detected outputs.

3.3.7.3.1 Sum-Channel Output - The sum-channel IF signal shall be detected and supplied to a video amplifier that shall provide a video output of 2 volts ±10% peak across the load of the system. The amplifier shall be capable of passing video pulses from 0.5 to 1000 microseconds duration with a maximum rise time of 0.25 microseconds and a maximum droop of 1.0 db. The input-output amplitude response shall be linear within 25% between 5% and 150% of normal output voltage. Video noise and hum generated within the video amplifier shall be less than 1% of normal output voltage.

3.3.7.3.2 Monopulse Output - The difference channel IF signal shall be compared with the sum-channel IF signal in a phase comparator circuit. The resultant bipolar video after amplification shall provide an output between plus 2 volts ±10% peak and -2 volts ±10% peak across the load of the system. The input-output amplitude response shall be linear within 25% between 5% and 150% of normal output voltage. Video noise and hum generated within the video amplifier shall be less than 1% of normal output voltage.

3.3.7.4 AFC Subassembly - The AFC subassembly shall receive an IF sample from the AFC mixer and shall furnish a modulating signal capable of holding the local oscillator to a frequency 30 ±0.25 mcs above the transmitter frequency at all times.

3.3.7.5 Local Oscillator Subassembly - The local oscillator subassembly shall contain the local oscillator and a variable attenuator to control crystal current. There shall be three isolated outputs provided for the two signal balanced mixers and the AFC mixer.

3.3.7.5.1 Local Oscillator - The local oscillator shall be capable of generating a frequency 30 mcs above the transmitter frequency by means of a modulating signal from the AFC subassembly.

3.3.7.6 Receiver Control Subassemblies - The receiver control subassemblies shall provide heater power for the receiver, STC circuits, gated AGC circuits, and keep alive voltage for the T-R tubes.

3.3.7.6.1 Sensitivity Time Control - The STC shall be capable of reducing the IF pre-amplifier sensitivity of the sum-channel and the difference-channel by a minimum of 70 db at five miles. Depth of drive and recovery shall be adjustable, each by means of one operating control common to the sum-channel and difference-channel IF preamplifiers located on the Radar Set Control. With STC adjusted to 70 db depth of drive at 5 miles, the recovery shall be adjustable to make the sensitivity 3 db below maximum at 50 to 150 miles.

3.3.7.6.2 Gated Automatic Gain Control - A gated AGC voltage shall be generated utilizing the radar video from the sum-channel video amplifier and a gate of nominal 50 microseconds width from the IP(*)/APS-82. The gated AGC voltage shall be furnished to the sum-channel and the difference-channel IF preamplifiers, to control their gain to prevent saturation during the gate width.

3.3.8 Uni-Directional Coupler CU-(*)/APS-82

3.3.8.1 Function - The Uni-Directional Coupler shall provide for coupling a fixed percentage of an auxiliary line power into the difference-channel (directed towards receiver).

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3.3.8.2 VSWR - The VSWR of the Uni-Directional Coupler shall not exceed 1.12.

3.3.8.3 Coupling - The coupling attenuation shall be 40 ±2 db, and the output jack shall be marked with the attenuation to the nearest 1/10 db. The coupling shall be flat within ±0.25 db over the band of 2850 to 2910 mc/s.

3.3.8.4 Directivity - The directivity shall be a minimum of 20 db.

3.3.8.5 Electrical Connections - The following electrical connections shall be provided:

(1) Waveguide to Monoplexer

(2) Waveguide to Antenna

(3) Test Jack

3.3.9 Monoplexer Assembly CU-(*)/APS-82

3.3.9.1 Function - The monoplexer assembly shall interconnect the antenna difference channel and the difference receiver.

3.3.9.2 Contents of Unit - The monoplexer assembly shall consist of the following parts:

3.3.9.2.1 TR Tube - A TR tube shall be coupled to the antenna difference-channel. The TR tube shall be identical to that of 3.3.5.2.2.

3.3.9.2.2 Crystal Protector - The crystal protector shall consist of a solenoid-operated device which will automatically cause an insertion loss of at least 35 db at all frequencies (measured between matched waveguide terminations) whenever keep-alive voltage is not applied to the TR tube. The crystal protector shall have waveguide flanges for insertion between TR tube (3.3.9.2.1) and image rejection filter (3.3.9.2.3) or balanced mixer (3.3.9.2.4).

3.3.9.2.2.1 The solenoid shall be capable of being continuously energized under any combination of environmental conditions and shall consume no more than 2 amperes at 28 volts dc.

3.3.9.2.3 Image Rejection Filter - The image rejection filter, if used shall have a performance as specified in 3.3.5.2.4. It may be a separate item identical to that of 3.3.5.2.4 or it may be combined with the crystal protector (3.3.9.2.2).

3.3.9.2.4 Balanced Mixer - Shall be identical to that specified in 3.3.5.2.5.

3.3.9.2.5 I.F. Preamplifier - The I.F. preamplifier shall be identical to that specified in 3.3.5.2.6.

3.3.9.3 Controls - The following control shall be provided in the monoplexer when the image rejection filter is used:

IMAGE REJECTION TUNE

3.3.9.4 Electrical Connections - The following electrical connections shall be provided as described in 3.3.9.2:

(1) WAVEGUIDE TO UNI-DIRECTIONAL COUPLER

(2) KEEP-ALIVE

(3) I.F. OUTPUT

(4) CRYSTAL #1 CURRENT

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(5) CRYSTAL #2 CURRENT

(6) GROUND

(7) HEATER

(8) B+

(9) STC

(10) AGC

3.3.9.5 Loss - The monoplexer loss exclusive of mixer, shall not exceed 1.5 db.

3.3.9.6 VSWR - The monoplexer VSWR, shall not exceed 2:1.

3.3.10 Indicators, Azimuth and Range - IP-414/APA-125 - Two IP-414/APA-125 indicators shall be supplied. The indicators shall be identical to the Azimuth and Range Indicators of Specification MIL-I-19165.

3.3.10.1 Function - These units shall provide for displaying radar range and azimuth and AIC information.

3.3.11 Displays, Height Finder, IP-(*)/APS-82(2) -

3.3.11.1 Function - The units shall provide a visual indication for the centering the antenna on a target in the elevation plane. Each unit shall include an altitude computer and automatic gating circuitry for examining the target in range and azimuth.

3.3.11.2 Contents of Units - Each of the two units shall contain the following:

3.3.11.2.1 Azimuth Gate Generator - The azimuth gate generator shall produce two gates, the narrow azimuth gate and the wide azimuth gate.

3.3.11.2.1.1 Narrow Azimuth Gate - The narrow azimuth gate shall be less than 10 degrees in width. It shall be used for gating the video display of the Height Finder Display and for defining the position of the gate on the Range and Azimuth Indicator.

3.3.11.2.1.2 Wide Azimuth Gate - The wide azimuth gate shall be approximately 40 degrees wide. The leading edge of the wide azimuth gate shall precede the narrow azimuth gate by approximately 30 degrees. It shall be used to furnish speed control data to the antenna drive system.

3.3.11.2.1.3 Azimuth Gate Position Control - The position of the azimuth gates shall be determined by the setting of Cursor #1 Bearing Control on the associated Range and Azimuth Indicator, IP-414/APA-125.

3.3.11.2.2. Range Gate Generator - The range gate generator shall produce a range gate of 50 ±25 microseconds to be used for gating signals and sweep on the Height Finder Display.

3.3.11.2.2.1 Range Gate Position Control - The position of the range gate shall be determined by the setting of the "Miles - Strobe Range 1" control on the Associated Range and Azimuth Indicator IP-414/APA-125.

3.3.11.2.3 Height Finder Presentation -

3.3.11.2.3.1 Cathode Ray Tube - A three-inch, cathode ray tube shall be used in the Height Indicator.

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3.3.11.2.3.2 Video Amplifiers - Video amplifiers shall be used to provide deflection of the cathode ray tube. The amplifiers shall be capable of passing video pulses from 0.5 to 1000 microseconds duration with a maximum rise time of 0.25 microsecond and a maximum droop of 1.0 db. The input-output amplitude response shall be linear within 25% between 5% and 150% of normal output voltage. Video noise and hum generated within the video amplifier shall be less than 1% of normal output voltage.

3.3.11.2.3.3 Sweep Amplifier - A sweep amplifier shall be used to produce 50 ±25 microsecond sweeps on the face of the cathode ray tube.

3.3.11.2.3.4 Sum Video Display - The sum video shall be displayed as a displacement of the sweep.

3.3.11.2.3.5 Monopulse Video Display - The monopulse video shall be used to indicate whether the target is above or below the antenna beam center.

3.3.11.2.4 Height Computer - The indicator shall contain a computer capable of producing a dc voltage proportional to the sum of the following terms:

(1) $R^2/2r$, where R is the slant radar range and r is the radar radius of the earth.

(2) R Sine E where E is the elevation angle of the antenna beam.

(3) h_A which is the altitude of the AEW aircraft.

3.3.11.2.4.1 Indication - The output of the height computer shall indicate target height in thousands of feet from zero to 99,000 feet above sea level.

3.3.11.2.4.2 AEW Altitude Control - A front panel control, calibrated in thousands of feet, shall be provided, to manually insert AEW aircraft height from zero to 25,000 feet.

3.3.11.2.4.3 Antenna Tilt Control - A front panel control for manually controlling the tilt of the antenna during the height finding phase shall be provided.

3.3.11.2.5 Antenna Speed Control Relay - A signal shall be provided for controlling the antenna azimuth speed. It shall be controlled by the leading edge of the wide azimuth gate and the trailing edge of the narrow azimuth gate.

3.3.11.2.6 Additional Controls - Additional front panel controls shall be provided as follows:

(1) Video Gain

(2) Ht. Fnd.-Out Switch

(3) Height Finding Indicating Lights

(4) Tilt Limit Indicator Light

3.3.11.2.7 Information to IP-414/APA-125 - During the height finding operation, the area being searched shall be defined by two arcs corresponding to the leading and trailing edges of the range gate. This information shall be supplied in the form of video signals for mixing and display on the Range and Azimuth Indicators.

3.3.12 Controls, A.I.C., C-(*)/APS-82.

3.3.12.1 Function - Two Air Intercept Computer (AIC) Control Units shall contain the controls necessary for the AIC Computers, CP-(*)/APS-82.

3.3.12.2 Form Factor and Mounting - The units shall be designed to attach by means of adjustable brackets to the Range and Azimuth Indicators IP-414/APA-125. The panel shall be capable of being hinged upward to allow clearance for the operator when he takes his position. The hinge mounting shall be easily detached from the indicator unit for maintenance purposes.

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3.3.12.3 Contents of Unit - Each of the two units shall contain the joystick assemblies, synchros, potentiometers, switches, etc., required to supply information to the AIC Computers CP-(*)/APS-82.

3.3.12.4 Controls - The following controls shall be available on the front panel of the AIC Control Units, and shall serve the indicated purpose.

3.3.12.4.1 "Interceptor Slew" - There shall be a two speed joystick to slew the origin of the interceptor cursor. The two speed push-button shall also reset the interceptor tracking time to zero.

3.3.12.4.2 "Target Slew" - There shall be a two speed joystick to slew the origin of the target cursor. The two speed push-button shall reset the target tracking time to zero.

3.3.12.4.3 "Timer" Switch - Applies voltage to start the basic tracking timer motor. In the off position, the tracking timer shall reset to zero.

3.3.12.4.4 "Target Bearing" - Provides bearing information for the Target cursor. The bearing accuracy shall be ±1% and markings shall be no more than 5 degrees apart. A second marker shall be provided on the dial to indicate reciprocal course.

3.3.12.4.5 "Interceptor Bearing" - Provides bearing information for the Interceptor cursor. Bearing markings shall be no more than 5 degrees apart, and two additional markers shall be provided on the dial to indicate orthogonal courses.

3.3.12.4.6 "Target Speed" - Provides target speed information for the AIC Computer. The markings shall be a maximum of 50 knots apart, and shall cover the speed range of 100 to 1,000 knots.

3.3.12.4.7 "Interceptor Speed" - Provides interceptor speed information for the AIC Computer. The markings shall be a maximum of 50 knots apart, and shall cover the speed range of 100 to 1,000 knots.

3.3.12.4.8 "Time, Fwd-F-S, Rev. -S-F" - A switch which allows the advance time to be driven in a forward or a reverse direction at a fast or slow speed. It shall be possible to advance time to 30 minutes in no more than 15 seconds in the fast speed position.

3.3.12.4.9 "Offset Distance" - Provides distance information for the offset computer. The calibration shall be in 1 mile increments and cover a range of 10 to 50 miles.

3.3.12.4.10 "Offset Angle" - Provides bearing information for the offset computer. The calibration marks shall be a minimum of 2 degrees apart and cover a range of ±60 degrees.

3.3.12.4.11 "Time to Go" Indicator - Provides information as to the time remaining to the point of interception on a collision course intercept or to the point of turn on an offset course intercept. This repeat indicator shall follow the timer with an accuracy of ±5 seconds. The smallest graduation shall be no more than 15 seconds.

3.3.13 Computers, A.I.C., CP-(*)/APS-82 (2)

3.3.13.1 Function - Two AIC Computer Units, in conjunction with the Range and Azimuth Indicators IP-414/APA-125 and the AIC Control Units C-(*)/APS-82, shall allow the operator to perform either collision or offset intercept computations. In addition, the units shall contain the servo amplifiers for controlling the follow-up units in the IP-414/APA-125.

3.3.13.2 Contents of Unit - Each of the two AIC Computers shall contain the following:

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3.3.13.2.1 Range Strobe Generator - One each Range Strobe Generator shall be provided for the Interceptor cursor and the Target cursor. The Range Strobe Generator shall provide output pulses which, when controlled by the velocity and time controls, will indicate the expected position of the target at the present time and at some future time. The maximum range of the strobe generator shall be 500 miles, using 6.18μ seconds per mile. The accuracy of the range strobe shall be ±1 mile or ±1% whichever permits the greatest error.

3.3.13.2.2 Servo Amplifiers - The AIC unit shall house the three servo amplifiers associated with the operation of the Range and Azimuth Indicator IP-414/APA-125.

3.3.13.2.3 Timing Mechanism - The timing mechanism shall contain the following:

3.3.13.2.3.1 Tracking Potentiometers - Potentiometers shall be provided for controlling the delay of the Range Strobe Generators of paragraph 3.3.13.2.1. The wiper arms of these potentiometers shall be rotated at a speed compatable with paragraph 3.3.12.4.8.

3.3.13.2.3.2 Advance Time Potentiometers - Potentiometers shall be provided for controlling the advance delay of the Range Strobe Generators. The total time required for total advance of the wiper arm of the potentiometer shall be a minimum of 15 seconds.

3.3.13.2.3.3 Reset - Provisions shall be made for resetting the potentiometers in less than 2 seconds.

3.3.13.2.3.4 Limits - Mechanical stops shall be provided to limit the rotation of the potentiometers in either the forward or reverse direction. The stops shall be such that the mechanical stops of the potentiometers are not used.

3.3.13.2.3.5 Timer - Timing information shall be provided by an ac synchronous motor with a suitable speed reduction. The accuracy of the timer shall be directly proportional to the power line frequency.

3.3.13.3 Controls - The following controls shall be provided on the front panel of the AIC Computer.

3.3.13.3.1 "Combined Slew" Joystick - A two speed geared joystick shall be provided to allow combined slew of the origins of the sweep and cursors together.

3.3.13.3.2 "Combined Slew, In-Out" Switch - A front panel switch shall be provided to disable the combined slew function.

3.3.13.3.3 "Fleet Center" Controls - A two speed joystick shall be provided as front panel controls to provide for centering the origin of cursor #2 during the search operation.

3.3.13.3.4 "Fleet Center Slew, In-Out" Switch - A front panel switch shall be provided to disable the fleet center slew function.

3.3.13.3.5 "RGS In-Out" Switch - A front panel switch shall be provided to allow selection of the Radar Ground Stabilization Units.

3.3.13.3.6 Function Switch, "Search- Int.-Off-Set Int." - A front panel switch shall be provided to select the various modes of operation.

3.3.14 Amplifier, Electronic Control AM-(*)/APS-82

3.3.14.1 Function - The Electronic Control Amplifier shall receive the tilt error signal, process this signal, and utilize this information to furnish power for the antenna tilt drive motor. In addition, such derivative and integral control as may be required to meet this specification shall be incorporated.

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3.3.14.2 Contents of Units - The Electronic Control Amplifier shall contain the following:

(1) Servo Amplifier

(2) Magnetic Amplifier

(3) Speed changing control relays

3.3.14.2.1 Servo Amplifier - A servo amplifier shall be provided to process the tilt angle information.

3.3.14.2.2 Magnetic Amplifier - A magnetic amplifier shall be provided to provide power for the antenna tilt drive motor.

3.3.14.2.3 Speed Changing Relays - Relays shall be provided for changing the antenna azimuth speed during height finding and selecting the proper tilt control.

3.3.15 Power Supply PP-(*)/APS-82

3.3.15.1 Function - This unit shall furnish regulated and unregulated voltages to the component units comprising this radar system. It shall also furnish regulated voltages to two Radar Ground Stabilization units of the AN/ASA-13( ).

3.3.15.2 Output Characteristics - The following voltages shall be made available at the output connectors.

Regulation	Nominal Volts	DC Regulation Maximum Allowable Voltage Variation in Volts (±)	Max. Peak to Peak Ripple in Millivolts under System Load	Approx. Current in ma supplied to System
Unregulated	+300 DC ±30	9	200	750
Regulated	+250 DC ±25	.25	100	900
Regulated	+150 DC ±1.5	.15	50	925
Regulated	-150 DC ±1.5	.15	50	850
Regulated	-150 DC ±3.0	.30	50	135

3.3.15.2.1 Regulation - Regulated output voltages shall be held within specified limits for loads from 10% to 110% of normal current.

3.3.15.2.2 Heading Marker Generator - The power supply shall contain the heading marker generator to supply a video signal consisting of approximately 80 KC pulses for a duration of approximately 8 millisecs. The generation of these pulses is determined by the closing of a microswitch in the antenna pedestal as described in paragraph 3.3.1.18.8 above. The duration of the heading marks shall be independent of the opening of this switch.

3.3.16 Radar Set Control C-(*)/APS-82

3.3.16.1 Function - The radar control shall provide for the main power switch, control of antenna rotation and manual tilt, receiver STC slope, STC depth of drive control, transmitter control, one each warning and standby lights, and magnetron current indicator.

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3.3.16.2 Controls -The following controls shall be provided:

(1) The main power control shall be a three-position Off-Standby-Operate switch that controls contactors in the aircraft distribution box to provide the following. In the Standby position, it shall turn on all the filaments, apply keep-alive voltages, energize the various blowers and pressurization unit, apply voltages from the low voltage power supply to the system. In the Operate position, it shall provide all operating voltages to the system provided that the time delay relay has timed out.

(2) Receiver - STC slope control.

(3) STC depth of drive control.

(4) Antenna Azimuth - STOW-STOP-ROT Switch - In the STOW position it shall provide for stopping the azimuth mechanism in a discreet position. In the ROT position, the antenna shall rotate continuously at its normal speeds. In the STOP position, the power shall be removed from the azimuth drive motor but the unit shall not stow.

(5) Antenna Tilt Control - This control shall provide for manually adjusting the antenna beam tilt over the tilt range and shall indicate the beam tilt over a range of ±15 degrees about the horizontal axis except during height finding. The tilt angle shall be marked on the panel in one degree increments.

(6) UP-DOWN Switch - A three-position UP-DOWN switch shall be provided to enable manual adjustment of the magnetron current in the transmitter (see 3.3.16.5).

3.3.16.3 Warning Lights - Two warning lights shall be provided that light under the following system conditions.

(1) When the main switch is in the STANDBY position or in the Operate position before the time delay has timed out.

(2) When an overload or overheating has occurred in the transmitter, or when insufficient air pressure exists in the waveguides.

3.3.16.4 Plastic Front Panel - The front panel of this unit shall be a separate plastic lighting sheet per Specification MIL-P-7788, except that the lighting shall be white. Function and position of all controls shall be visible under all conditions of illumination, except that it will not be required to indicate the position of toggle switches. Provision shall be made for external control of the intensity of the lighting.

3.3.16.5 Magnetron Current Indicator - A meter shall be provided which by its indication, shall permit proper magnetron setting by means of the UP-DOWN switch. The scale shall show the region of proper setting, and there shall be no requirement for calibration in terms of magnetron current.

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3.3.17 Maintenance Control C-(*)/APS-82

3.3.17.1 Function - The maintenance control shall provide means for monitoring voltages, currents and significant waveforms of the system. It shall provide also for receiver control and magnetron current control.

3.3.17.2 Controls - The following controls shall be provided:

(1) Receiver-MAN-AFC switch.

(2) Receiver Manual tuning control.

(3) Receiver Radar Gain Control.

(4) Magnetron UP-DOWN switch. This switch shall enable control of magnetron current.

(5) EMERGENCY OFF push-button switch. Activation of this switch shall turn off the radar set and all electronic equipment tied into this system. Resetting shall be accomplished by switching the main switch on the radar set control to the OFF position.

(6) Receiver Crystal Current - FORWARD-REVERSE switch. In the FORWARD position, it shall enable the metering of the forward crystal currents. In the REVERSE position it shall permit displaying on the meter an indication of reverse crystal currents.

(7) Metering Switch - This switch shall select the circuits to be metered. They shall be as follows:

(a) Five crystal currents

(b) Keep alive current

(c) Magnetron current

(d) Modulator current

(e) Low Voltage Power Supply dc output voltages.

(8) Video Selector - This switch shall select the triggers and waveforms to be furnished to BNC jacks on the front panel. Waveforms and triggers available for display shall be as follows:

(a) Sum-channel video/Mod. Trigger

(b) Difference-channel video-Modulator Trigger

(c) Relay data/Basic Pulse

(d) Thyratron Trigger/Cosine Pulse

(e) Transmit Pulse/Cosine Pulse

(f) IFF Mode I/Mod. Trigger

(g) Relay Test Video/Basic Pulse

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(9) Normal-Transmitter Test Switch - In the NORMAL position, the waveguide switch shall connect the radar system to the antenna when the main switch is on STANDBY or OPERATE and shall connect the radar system to the dummy load when the main power switch is OFF. In the TRANSMITTER TEST Position, the radar system shall connect to the dummy load.

3.3.17.3 Monitor Meter - A monitoring meter shall be provided. The meter shall be capable of measuring the DC power supply voltages, high voltage power supply current, magnetron anode current, crystal currents, and keep-alive current. The meter shall be accurate to ±2% of full scale.

3.3.17.4 Mounting - This unit shall be mounted on Mounting Base MT-(*)/APS-82.

3.3.18 Transformer Assembly TF-(*)/APS-82

3.3.18.1 Function - The transformer assembly shall provide a motor-driven, three-phase variable autotransformer to enable control of the magnetron current from the Maintenance Set Control or Radar Set Control.

3.3.18.2 Rating - The autotransformer shall provide 100% voltage adjustment and be capable of delivering up to 5 amps line current.

3.3.19 Air Compressor HD-(*)/APS-82

3.3.19.1 Function - This unit shall provide and maintain pressurization of the r-f plumbing and the radar transmitter. The pump shall be capable of maintaining the pressurized r-f plumbing and the radar transmitter between 17 and 22 psi absolute under all operating conditions specified herein.

3.3.20 Dehydrator HD-(*)/APS-82

3.3.20.1 Function - The dehydrator shall contain an activated dessicant for maintaining dry air in the pressurized components of the system. The dehydrator shall be so constructed that, when filled with dessicant, and when the inlet and outlet are sealed, the dessicant will remain activated for a period of not less than 90 days.

3.3.20.2 Visual Indicator - A visual indicator shall be provided to aid in determining the condition of the dessicant, and attention shall be given to the ease of changing the dessicant.

3.3.21 Radar Relay Synchronizer SN-(*)/APS-82

3.3.21.1 Function - This unit shall accept modulator trigger, radar video, IFF video, IFF coding pulses and azimuth information from the radar system, and supply the IFF Trigger to the AN/APX-7. The radar relay synchronizer shall convert the azimuth information into azimuth sync pulses and shall be triggered by the modulator trigger. These azimuth sync pulses and radar and IFF video shall modulate the radar relay transmitter AN/ART-28( ). There shall also be alarm and AGC circuitry using externally sampled radar relay transmitted information to insure adequate transmission and a proper azimuth sync to video signal ratio.

3.3.21.2 Basic Pulse - The basic pulse shall follow the modulator trigger by a time given by the following equation:

$$t_{Basic} = \frac{1}{f} - 401 - t_1 \quad (\text{Sine } \theta + \text{Cos } \theta)$$

f = prf

t_1 = 109.5 usec

θ = the angle between the axis of the radar beam and true north

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The time delay error of the basic pulse shall be less than 4% of the constant t_1.

3.3.21.3 Sine Pulse - The sine pulse shall follow the basic pulse by a time given by the following equation:

$$t_{sine} = 185.7 + t_1 \quad \text{Sine } \theta$$

The time delay error of the sine pulse shall be less than 4% of the constant t_1.

3.3.21.4 Cosine Pulse - The cosine pulse shall follow the sine pulse by a time given by the following equation:

$$t_{cosine} = 185.7 + t_1 \quad \text{Cos } \theta$$

The time delay error of the cosine pulse shall be less than 4% of the constant t_1. The cosine pulse will be used as the IFF trigger and shall be of positive polarity and have a minimum peak amplitude of 5 volts as measured across a 500 μμf capacitance. The delay between the cosine pulse and the transmitter pulse shall be 32 μseconds and the pulse to pulse jitter shall not exceed ±2 μseconds.

3.3.21.5 Relay Azimuth Information - The modulating signal for the radar relay transmitter shall consist of basic, sine and cosine pulses, plus the modulator pulse or the coded IR pulse. Each pulse shall be transformed into three one microsecond pulses whose leading edges are separated by 3 ±0.3 and 5 ±0.5 microseconds respectively. This signal shall be of positive polarity with a peak voltage of 5 ±1 volt as measured across a 500 μμf capacitance.

3.3.21.6 IFF Mode Identification - The IFF mode identification data shall be composed of one of three IFF mode identification pulses (coded IR pulses) supplied by the AN/APX-7 IFF equipment. This signal will occur at 15, 20, or 25 microseconds after the cosine pulse depending upon the IFF mode of operation selected. The pulses shall be of positive polarity and have a minimum peak amplitude of 10 volts. Each IR pulse shall be transformed into three one microsecond pulses whose leading edges are separated by 3 ±0.3 and 5 ±0.5 microseconds respectively. During radar-IFF time sharing, the coded IR pulse will replace the modulator pulse on those intervals where only IFF video is relayed.

3.3.21.7 Relay Video - The video data shall be of positive polarity, and adjustable in amplitude from 0 to 2 volts measured across 500 μμf capacitance. It shall follow the synchronizing data and consist of radar and/or IFF video. During IFF interrogation the radar video shall alternate on successive cycles with the IFF video. In addition, provisions shall be incorporated to permit IFF video to be mixed directly with radar video by selection at the Relay Control Box.

3.3.21.8 Azimuth Servo - The azimuth servo shall be designed to follow true-bearing azimuth voltages from a one-speed, synchro differential generator in the Relay Synchronizer Control Box, and position a sine-cosine potentiometer. The d-c sine and cosine true-bearing azimuth voltages thus obtained shall control the time delay of basic, sine and cosine pulses. The servo dynamic error shall be less than 1/3 degree at an azimuth rotation of 6 revolutions per minute.

3.3.21.9 Automatic Gain Control - This circuit shall sample the transmitted r-f power from the AN/ART-28( ) and control the ratio between video and sync signals by means of the detected signal. An alarm shall give indication of low transmitted output.

3.3.21.10 Power Requirement - Two power sources are required for operation of radar relay synchronizer, three-phase, 200 VAC, line to line, 0.3 KVA and 28V d-c 1 amp. (see 3.2.10.1).

3.3.21.11 Mounting - This unit shall be mounted on Mounting Base MT-(*)/APS-82.

3.3.22 Relay Synchronizer Control Box C-(*)/APS-82

3.3.22.1 Function - The Relay Synchronizer Control Box shall control the primary power for the AN/ART-28( ), the IFF information to be relayed, and the compass variation control.

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3.3.22.2 Controls - The following controls shall be provided:

(1) Variation Set - The Variation Set permits manual input of the variation of 59° East or West so that the Relay radar information may be supplied with true North as a reference.

(2) Relay Power ON-OFF - This switch turns the AN/ART-28 on and off.

(3) IFF Out-Share-Mix - This switch allows the choice of relaying IFF interlaced with the radar video or mixed with the radar video.

3.3.22.3 Plastic Front Panel - The front panel of this unit shall be a separate plastic lighting sheet per Specification MIL-P-7788 except that the lighting shall be white. Function and position of all controls shall be visible under all conditions of illumination, except that it will not be required to indicate the position of toggle switches. Provision shall be made for external control of the lighting intensity.

4. QUALITY ASSURANCE PROVISIONS

4.1 Classification of Tests

(1) Preproduction Tests - Preproduction Tests are those tests conducted on an equipment representative of the equipments to be supplied under the contract, for the purpose of determining that the equipment meets all the requirements of the applicable specifications. Preproduction Tests shall be accomplished by Contractor Demonstration Tests and Service Approval Tests.

(a) Contractor Demonstration Tests - Contractor Demonstration Tests shall be accomplished under the responsibility of the contractor to demonstrate that the equipment complies with all applicable requirements.

(b) Service Approval Tests - Service Approval Tests shall be accomplished under the responsibility of the procuring service for the purpose of verifying the results of the Contractor's Demonstration Tests and for conducting any additional tests deemed necessary to determine that the equipment complies with all applicable requirements.

2) Acceptance Tests - Acceptance Tests are those tests accomplished on equipment submitted for acceptance under the contract. Acceptance Tests shall consist of the following:

(a) Individual Tests - Individual Tests are those tests conducted on each equipment submitted for acceptance under the contract.

(b) Sampling Tests - Sampling Tests are those tests accomplished on a quantity of equipments as approved by the Bureau of Aeronautics and which have previously passed individual tests.

(c) Special Tests - Special Tests are those tests accomplished on a quantity of equipments as approved by the Bureau of Aeronautics.

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(3) Life Tests - Life Tests shall be conducted by the Contractor at his plant. These tests shall be started as soon as practicable on an early production equipment.

4.2 Preproduction Tests -

4.2.1 Contractor Demonstration Tests - The contractor shall conduct tests on one or more equipments as necessary to determine that the contractor's proposed model will meet all the requirements of the applicable specifications and the contract. Contractor Demonstration Tests shall be conducted in accordance with the approved test procedure. The data obtained by the contractor in conducting these tests shall be included with the design data submitted with the equipment for Service Approval Tests. The Inspector of Naval Material and the Bureau of Aeronautics shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. When tests are supervised the Bureau of Aeronautics may, when conducting Service Approval Tests, omit all tests which will duplicate tests previously conducted by the contractor; however, in conducting Service Approval Tests, the Bureau of Aeronautics reserves the right to repeat any test previously conducted when deemed necessary. Contractors not having laboratory facilities to conduct all required tests shall obtain either the services of a commercial testing laboratory or written approval from the Bureau of Aeronautics to omit that portion of the tests which cannot be conducted.

4.2.2 Service Approval Tests - The Service Approval Sample shall consist of one equipment representative of the production equipment to be supplied under the contract. This equipment shall be delivered by the Contractor to the destination specified by the Bureau of Aeronautics.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for Service Approval Tests, the contractor shall also submit the accessory material and design and test data specified in MIL-E-5400. The design and test data shall indicate the physical and electrical characteristics of the equipment and establish that the equipment meets all applicable requirements. The test data shall include data obtained by the contractor in testing to MIL-T-5422.

4.2.3 Scope of Tests - Preproduction Tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction Tests shall include environmental tests in accordance with MIL-T-5422 and interference tests in accordance with MIL-I-6181. The sampling and testing of plastic lighting plates shall be in accordance with MIL-P-7788.

4.2.4 Preproduction Approval - Approval of the preproduction sample shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No additional equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of any equipments prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance under the contract. The preproduction sample shall not be considered as one of the equipments under the contract, however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the required Acceptance Tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having laboratory testing facilities satisfactory to the Bureau of Aeronautics shall engage the services of a commercial testing laboratory acceptable to the Bureau of Aeronautics. The contractor shall furnish test reports showing quantitative results for all Acceptance Tests required by this specification and signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guarantee of the acceptance of the finished product. Acceptance Tests shall consist of the following:

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(1) Individual Tests - Individual Tests shall be conducted on each equipment submitted for acceptance under the contract.

(2) Sampling Tests - Sampling Tests shall be conducted on a quantity of equipments as specified in 4.3.2.1. Equipments for sampling tests shall be selected by the Government Inspector after the equipment has passed the Individual Tests.

(3) Special Tests - Special Tests shall be conducted on a quantity of equipment as specified in paragraph 4.3.3.1.

4.3.1 Individual Test - Each equipment submitted for acceptance shall be subjected to the individual test to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum each equipment accepted shall successfully pass the tests specified in the following paragraph.

4.3.1.1 Scope of Tests - As a minimum, each equipment shall be subjected to the following:

(1) Examination of Product

(2) Operational Test

(3) Reliability Test

4.3.1.1.1 Examination of Products - Each equipment shall be carefully examined to determine that the requirements for material and workmanship as specified in MIL-E-5400 has been met. Also, it shall be determined that approval has been granted for the use of all non-standard parts.

4.3.1.1.2 Operational Test - Each equipment shall be operated. The period of operation shall be long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.1.1.3 Reliability Test - Each equipment shall be operated (without isolators) under the conditions specified below for a period of 12 hours without a failure. A failure shall be defined as anything which causes malfunctioning of the equipment. Only such adjustments shall be permitted which can be made using such controls and adjustments which would normally be accessible during flight.

Temperature	Ambient Room
Humidity	Ambient Room
Vibration (10 minutes)	Any frequency within the range of 20 to 60 cps (excluding resonant points) and an amplitude of ±3 g's

A 10-minute vibration run shall be made at the beginning of the 12-hour period. The equipment shall be nonoperating during the vibration run. During the 12-hour period of operation, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it shall be repaired and the test started over, except the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 12-hour period as specified above may be composed of several short periods of three hours or more.

4.3.2 Sampling Test - Sampling Test shall be conducted on a quantity of equipment as specified in the following paragraph. Equipment selected shall first have passed all of the individual tests. Sampling Tests shall be conducted on complete equipments.

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4.3.2.1 Sampling Test Schedule - Sampling Tests shall be performed on the following quantity of equipment. Selection of equipment for tests shall be under the supervision of the Naval Inspector.

Quantity of Equipments Delivered for Acceptance	Quantity of Equipment to be tested
From the first 10	1
From 11 to 100	2
From 101 to 200	1
From 200 and above	1 for each additional 200 or fraction thereof

4.3.2.2 Scope of Test - As a minimum each equipment selected for sampling test shall be subject to the tests specified below.

Tests	
I	Conduct a complete operational test at ambient room conditions and nominal input voltages, making all necessary measurements to assure all applicable specification requirements have been met.
II	Repeat the above using the highest and lowest specified input voltages and frequencies.
III	Conduct the reliability test specified in paragraph 4.3.1.1.3 except the test duration shall be 50 hours and two failures shall be permitted.

4.3.3 Special Tests - Special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. Equipment selected for Special Test may be selected from equipment previously subjected to the Sampling Test.

4.3.3.1 Special Test Schedule - Special Tests shall be conducted on the first and every fifth equipment that successfully passes the Sampling Test. However, additional equipment may be selected when required by the following:

(1) After an engineering change

(2) Whenever failure reports or other information indicate additional tests are necessary.

4.3.3.2 Scope of Test - In conducting Special Tests, test procedures previously approved for the Preproduction and Sampling Tests shall be used where applicable. When not applicable the contractor shall prepare a test procedure and submit to the Bureau of Aeronautics for approval prior to conducting the tests. As a minimum the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports only such tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

(2) Vibration Test (45 minutes cycling - 15 minutes in each of three planes)

(3) Humidity Tests

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4.3.4 Equipment Failure - Should a failure occur during either Sampling or Special Test, the following corrective action shall be taken.

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Suggest corrective action for Bureau of Aeronautics approval designed to reduce the possibility of the same failure occurring in future tests.

(4) Where practical include a test in the Individual Test to check all equipments for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4 Life Tests - The equipment shall be subjected to Life Test at the contractor's plant under the following simulated service conditions:

Temperature	Normal room temperature
Altitude	Normal (0 to 5000 ft.)
Humidity	Room ambient
A.C. Voltage	115 ±5 volts (at lowest applicable frequency)
D.C. Voltage	26.5 ±2.5 volts

4.4.1 Test Data - The contractor shall keep a daily record as to the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures during this test, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) Part type number

(2) The circuit reference symbol-number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure.

In the event of a part failure the Naval Inspector shall be notified immediately. A report shall be submitted to the Bureau of Aeronautics upon completion of the test. In this report the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The Bureau of Aeronautics will review such proposals and determine whether they are acceptable.

4.4.2 Test Duration - The duration of the Life Test shall be three hundred (300) hours.

4.4.2.1 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically the equipment shall be turned on and off several times and put through all phases of its performance.

4.4.2.2 Performance Check - At approximately eight (8) hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the Bureau of Aeronautics.

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4.4.3 Life Test Schedule - The equipment to be tested shall be representative of the current production in all significant aspects of design and manufacture. It shall be selected by the Government Inspector from equipments which have passed the Individual Tests. The Life Test shall be performed on the following quantities of equipment unless otherwise stated in the contract or approved by the Bureau of Aeronautics.

Quantity of Equipments Delivered for Acceptance	Quantity of Equipment to be tested
First 25	1
Next 26 to 200	1
Next 201 to 500	1
Over 500	1 for each additional 500 or fraction thereof.

4.4.4 Reconditioning of Test Samples - Equipment which has been subjected to the Life Test shall be reconditioned as follows:

(1) On completion of the Life Test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring agency.

(2) After reworking, the contractor may submit the equipment for acceptance under the contract.

4.5 Test Procedures - The procedures used for conducting Preproduction Tests, Acceptance Tests and Life Tests shall be prepared by the contractor and sent to the Bureau of Aeronautics for approval. The right is reserved by the Bureau of Aeronautics or the Naval Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts the Bureau of Aeronautics will furnish the procedures to be used for testing.

4.6 Pre-Submission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements. The design data submitted by the contractor for the preproduction model shall indicate its physical and electrical characteristics and other necessary test data which will establish that the equipment meets all the requirements of this specification.

4.7 Rejection and Retest - Electronic equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring agency.

4.8 Equipment Changes - After approval of the preproduction model, no changes shall be incorporated in the equipment unless approved by the Bureau of Aeronautics. To obtain Bureau of Aeronautics approval the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This applies for either a Bureau of Aeronautics' recommended change or a contractor's proposed change. When required the contractor shall also prepare an Electronic Material Change (EMC) and an Electronic Material Bulletin (EMB) to cover the approved change. ECP's, EMC's and EMB's shall be prepared in accordance with the requirements of SAR-300.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be packaged, packed and marked for shipment in accordance with the requirements of Specification MIL-P-17555 and MIL-STD-129.

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ±0.5 V, DC and 115/200 ±1V, AC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

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